

Module Title

Fundamentals of Data Science

Library Resource Management System

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Work Contribution and Reflective Statement

1. Work Contribution

My part of the Library Resource Management System was main.py, the file that ties the whole program together. I built the login loop, which asks for a username and password and gives the user up to three attempts before the program exits, and I built the two role-based menus, one for the Librarian and one for the Student that route the user to the correct set of options once they are logged in.

Once Sumant had the file-reading functions in data_io.py working, and Aman had the User, Librarian, and Student classes built in models.py, my job was to connect the two: main.py calls User.authenticate() to check a login, and then, depending on whether the object returned is a Librarian or a Student, it prints the matching menu and calls the correct methods when the user picks an option. I also wrapped every menu action in a try/except block so that a mistake, like typing an ISBN that does not exist, produces a clear message instead of crashing the whole program.

2. Reflection

What I did well

I am fairly new to Python classes, so at the start of this project I found it a bit intimidating that my file had to work with objects created in someone else's file. Once I understood that main.py did not need to know how authenticate() worked internally, only that it would either return a working user or raise an AuthenticationError, it became much easier. I think the menu system I built is clear and easy to follow, and testing it by deliberately typing wrong passwords and invalid menu options helped me see that the exception handling actually works the way it is supposed to.

Challenges I faced

The hardest part for me was understanding how Python passes an object between files. I had used functions before, but calling a method on an object that another file created, like librarian.add_book(...), was new. I also had to be careful with my loops in the menu functions, since an early version of my code exited the whole program after one action instead of showing the menu again.

What I learned

This project helped me understand why object-oriented programming is useful in a real system rather than just as a topic. Splitting the program into a data layer, a class layer, and a menu layer made it possible for three of us to work on the same project without constantly overwriting each other's code, as long as we agreed on the file formats and method names in advance. I also became a lot more comfortable with try/except, since I now understand it as a way of predicting what could realistically go wrong for a user, rather than just a syntax rule to memorise.

Working as a group

The biggest challenge for our group was making sure the three files worked together properly. At the beginning, we agreed on the format of the dictionaries returned by data_io.py. This helped Sumant, Aman, and me work on our own files separately. When we combined our work, we only had a few small problems, mainly making sure the method names matched exactly.